

Photon-Energy Transfer and Companion Deposition

A phenomenological framework for redshift, additional gravity and reservoir emission in a nonexpanding universe

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Abstract

We assess a hypothetical nonexpanding universe in which photons transfer energy to traveling companions that can enter gravitationally active reservoirs. The shared exact-third prescription improves galaxy rotation over ordinary matter alone, but the tested MOND prescription performs better. Freely fitted companion profiles describe six lens galaxies more closely than matched free NFW profiles; their free normalization no longer tests the one-third law or radiation supply. Six Coma shear bins do not distinguish the fitted companion and NFW shapes. A nearby distance-redshift fit gives a conversion gradient $c \alpha = 74.62$ km/s/Mpc, versus $H_0 = 74.92$ for the specified expansion comparator on the same inputs, with indistinguishable residuals at the precision of this diagnostic. A conditional supernova-brightness calibration instead favors 70.48 and retains redshift-dependent residuals. Recent calculations show that an extreme reservoir is not required by the inner data and that circular particle orbits can support the tested profiles in a frozen potential; formation and collective stability remain unresolved. These are useful partial results, not a common derived explanation of redshift, timing, lensing, supply and persistence. Known mathematics, project postulates and fitted rules are labeled separately.

1. Motivation and scientific scope

The objective is to ask whether energy originating in radiation can account for some phenomena conventionally attributed to additional gravitating matter, while a related propagation process accounts for redshift without metric expansion. The nonexpanding universe is a stipulated setting of the model, not an observational conclusion of this paper. Observed spectra, stellar motions, lensing images and radiation remain constraints to explain. No particular universe age or total size is imposed.

The absence of an identified dark-matter particle motivates alternatives but is not evidence for this specific mechanism. For example, the LZ collaboration's December 2025 analysis reported no WIMP signal in its stated search range [1]. That result constrains particle properties rather than erasing gravitational evidence. Our companions are likewise unobserved. Calling them radiation-derived rather than primordial does not exempt them from the evidential burden placed on an additional gravitating component.

We distinguish three comparisons: ordinary-matter Newtonian dynamics as a control; specified MOND and dark-halo models as gravitational competitors; and expanding cosmology as a separate propagation and large-scale benchmark. A successful galaxy fit alone cannot establish superiority over all three. Published distances may be adopted as scenario inputs, but their calibration and any dependence on the competing model must be recorded. A distance inferred from the same redshift law under test cannot serve as independent validation of that law.

2. Core postulates and unresolved choices

- P1. Photons progressively transfer energy into a companion sector. A surviving photon can have lower energy; simple removal of whole photons is insufficient.
- P2. Traveling companions are taken to propagate at light speed and retain their energy until an interaction. Their microscopic identity, polarization and relation to gravitons remain unspecified.
- P3. Capture transfers energy and momentum into a bound reservoir. The reservoir adds gravity under the reference effective-density prescription. Ordinary matter continues to gravitate.
- P4. The calibrated retention fraction has an exponent fixed to one-third. It is an empirical organizing rule, not a derived fundamental constant.
- P5. A captured reservoir may redistribute internally. Movement to deeper binding releases energy that must be accounted for together with support, heat and outgoing flux.
- P6. Companion-powered emission in black-hole environments is an optional branch. Evaporation of stored rest energy and release of settling energy are distinct mechanisms and must not be counted together as the same available budget.

An environmental time field remains a possible cause of P1, but no unique environmental clock law is established. Low gravity, absolute potential, density, curvature and tidal field are different candidate environmental variables. Substituting one for another changes predictions and requires an explicit choice. A freely chosen time coordinate by itself is not an observable interaction.

3. Minimal equations and provenance

3.1 Photon energy and redshift

Assume a local fractional energy-loss coefficient α per unit path length. With the standard photon energy-frequency relation, the deterministic reference law is

$$\frac{dE_\gamma}{d\ell} = -\alpha E_\gamma, \quad 1 + z = \frac{E_{\text{emit}}}{E_{\text{obs}}} = \exp\left(\int \alpha d\ell\right). \quad (1)$$

Provenance: the exponential solution is known mathematics and appears in other redshift models. Our proposed novelty would be an independently predictive physical equation for α , not equation (1). The archived illustrative constant is 0.0002488993 per Mpc; it does not establish broad-spectrum or independent distance-redshift agreement. At 100 million light-years it gives z approximately 0.00766 for the propagation component alone. Source/observer motions and ordinary endpoint gravitational effects must be treated separately.

An arrival-time map is also required:

$$t_o = F(t_e, \text{path}), \quad \frac{\Delta t_o}{\Delta t_e} \approx \frac{\partial F}{\partial t_e}. \quad (2)$$

Provenance: a general timing identity. Equation (1) does not determine equation (2). A static delay common to both ends of an event does not stretch its duration. The observed supernova width-redshift relationship [2] must be reproduced without using it merely to assign a new arbitrary factor. Different responses of electromagnetic and gravitational signals must also meet the GW170817 timing constraint [3], including

source emission-delay uncertainty.

3.2 Transport and capture

For energy densities u and energy fluxes F , a minimal conversion/capture ledger is

$$\partial_t u_Y + \nabla \cdot \mathbf{F}_Y = j_* - Q_{Yc}, \quad \partial_t u_c + \nabla \cdot \mathbf{F}_c = Q_{Yc} - C. \quad (3)$$

$$\partial_t u_d + \nabla \cdot \mathbf{F}_d = C - S_d. \quad (4)$$

Provenance: standard continuity equations; identifying the exchange terms with companions is proposed. S_d represents an actual energy transfer out of the deposited component and must enter another component or a boundary flux. If settling does gravitational work, kinetic and field-energy terms must be included; these three equations alone are not the full ledger. In a relativistic completion the total stress-energy exchanges must balance. This is not a claim that general relativity has a unique local gravitational energy density.

3.3 Retention and the reference deposited profile

$$X = \frac{L_{3.6}/(10^9 L_\odot)}{(R_d/\text{kpc})^2}, \quad \eta(X) = \frac{X^{1/3}}{1 + X^{1/3}}. \quad (5)$$

Provenance: the Hill-function form is established. Its radiation-density proxy, exponent choice and capture interpretation are project hypotheses. Here L refers to the adopted 3.6-micron luminosity proxy, not a measured total historical incident energy.

$$a = q_a R_d, \quad \kappa(r) = \frac{k_0}{[1 + (r/a)^2]^2}, \quad J(r) = \frac{1}{2} \int_{-1}^1 e^{-\tau(r, \mu)} d\mu. \quad (6)$$

The optical depth τ is the integral of κ along the incoming ray. The effective stored density is

$$\rho_d(r) = \frac{2C_0 \eta(X) J(r)}{[1 + (r/a)^2]^2}, \quad M_d(< r) = 4\pi \int_0^r \rho_d(x) x^2 dx. \quad (7)$$

Provenance: ray attenuation, angular averaging and spherical integration are known. The opacity shape and its stored-density interpretation are proposed. The current fitted constants are $C_0=4.72858924e7$ solar masses/kpc cubed, $k_0=0.2039029004/\text{kpc}$ and $q_a=2.770766589$. C_0 is accumulated effective normalization, not a measured companion supply or universe age. The capture radius factor q_a differs from the contraction factor s below.

$$v_c^2(r) = v_b^2(r) + \frac{GM_d(< r)}{r}. \quad (8)$$

Provenance: standard Newtonian spherical-source gravity with the proposed effective mass inserted. Equating a deposited energy density to mass density via energy divided by c squared is only a nonrelativistic reference closure. Traveling radiation, pressure and field stress require a more complete treatment. No adjustable energy-to-gravity amplification is derived here.

4. Optional physical branches retained for investigation

4.1 Phase-dependent settling and its energy outlet

The earlier phase-selector pilot uses an ideal homogeneous Bose-gas criterion:

$$T = \frac{m\sigma^2}{k_B}, \quad T_c = \frac{2\pi\hbar^2}{mk_B} \left[\frac{\rho_d/m}{\zeta(3/2)} \right]^{2/3}, \quad f = \max[0, 1 - (T/T_c)^{3/2}]. \quad (9)$$

Provenance: known ideal-gas statistics, proposed companion application. The adopted kinetic scale is G times enclosed monopole mass divided by 3r; it is not a measured temperature. Fraction f of each original shell is moved to sr and the rest stays. A shared inner-bin fit gives m=17.78 eV/c squared and s=0.8032. These values are effective fit parameters. Massive bound states would have to differ from the original light-speed traveling excitations; thermalization and particle-number behavior are unresolved. Published superfluid models motivate investigating phases [4] but do not derive this recipe.

With companion self-energy U_d , external-potential energy U_b and necessary global support energy K, the cooling requirement is

$$E_{\text{mech}} = K + U_b + U_d, \quad Q_{\text{settle}} = E_{\text{mech},i} - E_{\text{mech},f}. \quad (10)$$

Provenance: standard energy accounting. U_d includes the self-energy factor one-half; U_b does not. K is assigned from the scalar force virial, a necessary global condition rather than a local equilibrium or stability proof. The tested shared model releases about 4.5e50 J, or about 1.2e-8 of deposited rest energy. A fully cooled solution requires this energy to leave the bound subsystem. Positive energy release alone does not determine the fitted stopping radius. Later coupled circular-endpoint calculations include redistribution of angular momentum to a 30-60 kpc receiving region. Their all-bin RMS changes from 6.34/10.91 to 6.28/8.10 km/s under the two baselines, with release 4.47e50/4.43e50 J. These are endpoint constructions, not evolutionary or collective-stability proofs. Local pressure and saturation alternatives have not supplied a demonstrated joint solution.

4.2 Refraction and relativistic completion

$$\nabla \cdot [\varepsilon(\rho)\nabla\Phi] = 4\pi G(\rho_b + \rho_d). \quad (11)$$

Provenance: density-dependent gravitational permittivity is an existing refracted-gravity approach [5]. Our companion-density driver and retained deposition profile are adaptations. Tested variants do not improve reserved outer Milky Way predictions, so refraction remains a secondary experimental branch. A nonrelativistic Phi does not alone determine light bending: a conventional weak-field metric involves a second potential Psi, and lensing probes their sum. No independent lensing multiplier is adopted.

4.3 Microscopic conversion and environmental time

Photon-graviton mixing in an external magnetic field is an established theoretical process [6], but its ordinary coupling is weak and stationary whole-photon conversion does not redshift surviving photons. A proposed inelastic interaction or time-dependent field must calculate frequency change, photon-number change, momentum transfer, angular scattering and timing together. QED-like and scalar/axion effective-field descriptions remain candidates; string-inspired fields are a source of possible interactions, not numerical predictions without specified couplings [7]. None currently derives equations (1), (5) and (7) together.

For an illustrative Poisson process removing fraction δ per event, mean event count Λ gives mean energy $E_0 \exp(-\Lambda \delta)$ and relative energy variance $\exp(\Lambda \delta^2) - 1$. These are known Poisson identities applied to a hypothesis. At the archived α and a 100-million-light-year path, an illustrative added relative linewidth of $1e-5$ requires per-event loss below $1.31e-8$ and at least roughly 582,000 events. This is a design example, not an observed exclusion limit.

4.4 Companion-powered nuclear emission

$$L_\gamma = f_\gamma P_{\text{release}}, \quad \dot{E}_{\text{res}} = P_{\text{capture}} - P_{\text{release}}. \quad (12)$$

Provenance: ordinary reservoir bookkeeping; attributing nuclear power to companions is our optional hypothesis. All released channels must be included. The radiated fraction f_γ is not the retention fraction η . Binding-energy release preserves constituent rest mass at Newtonian order; rest-energy evaporation changes the permanence postulate unless replenished. Observed black-hole-environment light is not established Hawking radiation, and no escape from inside a conventional horizon is assumed.

At a Sagittarius A* reference luminosity of $1e29$ W or less [8], pure photon output corresponds to about $1.77e-11$ solar masses/year of energy. The representative M87 core luminosity $2.7e35$ W [9] corresponds to $4.77e-5$ solar masses/year; its jet beaming, aperture and excluded mechanical power limit that comparison. These values reconstruct required throughput from observed brightness, rather than predicting brightness. Routing the whole Galactic settling budget through the faint Sgr A* reference channel would take about $1.43e14$ years. This ratio imposes no universe age: it instead demands an explicit routing, efficiency and activity-history model.

4.5 Transfer, angular scattering and rate consistency

Subsequent fixed-gap calculations separate the energy removed per interaction from the number of interactions. For independent small steps of energy Δ , the conditional relative energy variance is approximately $\Delta z/E_{\text{emit}}$ in the no-removal ladder. Maintaining an illustrative relative linewidth of $1e-5$ at $z=1$ for 2 eV input therefore requires Δ no larger than about $2e-10$ eV. Added photon-removal channels impose separate brightness constraints. These are design requirements under the stipulated stochastic process, not measured universal limits. The fractional-loss example above and the fixed-gap ladder are distinct mechanisms.

Finite spatial overlap can narrow scattering but also suppress its total rate. In the tested Gaussian heavy-store approximation, matching a scattering angle that decreases with apparent source size produces a rate proportional to inverse source-distance squared times $\exp[-(\text{distance}/\text{cutoff})^2]$. At fixed coupling and target density, the effect then saturates near the source. The favorable angular prescription cannot be combined with an unchanged long-path loss coefficient. A positive continuum of gaps alone also fails to flatten the earlier point-interaction color dependence; inverse-designed spectral responses remain hypotheses.

Source-observer geometry further distinguishes accumulated direction change from apparent image width. One fixed-angle optical example over 30.66 Mpc gives about 0.520 arcseconds apparent directional RMS and 5025 seconds mean geometric delay. A prescribed source-size-tracking kernel gives much smaller moments, but lacks a derived local cause and fails the unchanged-rate Gaussian closure. These are conditional moments, not image fits or gamma-ray timing predictions. A stationary delay distribution does not by itself provide redshift-proportional event dilation.

4.6 Reciprocal loading and finite protected storage

A candidate paired-mode interaction raises a store by Delta while transferring a quantum from energy E_H to $E_L=E_H-\Delta$. Its Hermitian conjugate permits reversal. In an incoherent, equally weighted mode-pair approximation, known bosonic factors give

$$A = \Gamma n_H(1 + n_L), \quad B = \Gamma n_L(1 + n_H), \quad r = \frac{B}{A}. \quad (13)$$

Provenance: standard ladder-operator factors applied to our proposed effective interaction. Forward preference requires $n_H > n_L$; smooth same-direction thermal occupations do not supply it. Maintained source and escape fields can sustain loading in the tested open-system closure, but their geometry, absolute coupling, recoil and connection to traveling companions are not derived. For a one-step optical route, a 2 eV input contributing 1e-8 eV processes 2e8 times its net assembly power, with most energy remaining in outgoing light rather than heat.

Normalized collective states do not provide a free N-squared first-excitation rate from an empty domain: the factor is N. Partly excited symmetric states can have stronger enhancement, but require preparation energy and enhance reverse transitions too. These are known collective-spin results [12], not new graviton physics. Repeated assembly can succeed despite a small probability per attempt, while requiring large recycled energy traffic; the representative unbiased ladder circulates about 200,000 times its net assembly energy.

Adding one finite protected level and a relaxation mode with occupation n gives, in the zero-protected-decay equilibrium of the tested model,

$$P_{\text{protected}} = \frac{n + 1}{n + 1 + rn}. \quad (14)$$

Provenance: a derived stationary result of our specified finite-state model. Collective size cancels from this occupancy, although it affects replenishment rates. Fifty-four stationary cases include reverse export and conserve energy. Nonzero leakage requires replenishment; zero leakage is a stipulated lifetime, not an explanation of permanence. One finite level also cannot accumulate unlimited energy.

When the full fitted Milky Way energy inventory is assigned to tiny single-excitation sites, the illustrative site/mode ratios become extremely large. A particular homogeneous escape closure requires protected lifetimes above roughly 1e32-1e41 years for its 120 kpc, 90%-protected examples. These are conditional required lifetimes, not cosmic-age constraints. Larger collective release quanta can ease mode crowding, but require a consistent assembly, reverse-transfer and protection mechanism. Thermal and nonthermal outgoing channels remain separate alternatives; energy in transit is not deposited energy.

4.7 A threshold interpretation of the one-third reference

The reference response has an exact representation as a capacity-weighted mixture of ordinary saturation laws. With positive threshold t and exponent q ,

$$\rho_q(t) = \frac{\sin(\pi q)}{\pi} \frac{t^{q-1}}{1 + 2t^q \cos(\pi q) + t^{2q}}. \quad (15)$$

$$\int_0^\infty \frac{X}{X+t} \rho_q(t) dt = \frac{X^q}{1+X^q}, \quad q = \frac{1}{3}. \quad (16)$$

Provenance: known positive fractional-response mixture mathematics [13]. Selecting this distribution from the desired curve is inverse design, not a first-principles derivation of one-third. Equilibrium occupied capacity is not automatically permanent capture efficiency. A physical model must derive the threshold weighting and connect it to the incident field and stored mass.

Truncating and renormalizing this distribution to thresholds from $1e-6$ to $1e6$ changes predicted rotation speeds by at most 0.640 km/s across the existing 149 galaxies without refitting. A narrower $1e-3$ to $1e3$ range changes some speeds by 7.03 km/s and worsens aggregate validation/test velocity RMSE. Broad cutoffs preserve the reference by construction; they neither establish the distribution's existence nor improve its main residuals.

4.8 Formation history and retention after fading

One explicit local kinetics is $df_t/dt = \lambda[X(1-f_t) - t f_t]$. From empty storage under constant X , its standard solution is

$$f_t(u) = \frac{X}{X+t} [1 - e^{-(X+t)u}], \quad u = \lambda \Delta t. \quad (17)$$

Provenance: known linear rate-equation solution; applying a shared λ and the selected threshold distribution to companions is hypothetical. The overall time scale is undetermined. Across the catalogue inputs, reaching 90% equilibrium takes dimensionless durations from 0.08478 to 102.462 . Shared-duration scans produce mixed score changes and do not select a new preferred law.

After individually stopping the source at that target, losing the first 10% of formed storage takes 4.92% – 41.79% of the formation time. This ratio is independent of λ and uses no universe age. A slower residual tail persists, but does not make the whole reservoir permanent. These controlled histories are not observations of the named galaxies and do not model continued distant illumination. Equilibrium requires balancing capture and release; a successful amplitude fit does not establish the required energy throughput.

4.9 Local capacity, external illumination and recycling

Write $A=2$ $C_0=9.45717848e7$ solar masses/kpc cubed for the unadjusted reference source amplitude, and $g(r)=[1+(r/a)^2]^{-2}$. An audit corrected earlier capacity and recycling inventories that had used C_0 where A was required. Equation (7) and the original lens calculations already contained the factor two. The correction doubles affected absolute mass inventories; the previously reported mass-ratio-based velocity predictions and scores are unchanged.

The tested alternative profiles are

$$\rho_{\text{ref}} = A\eta(X)Jg, \quad \rho_{\text{local}} = A\eta(XJ)g, \quad \rho_{\text{recycle}} = A\eta(X)g. \quad (18)$$

Provenance: the exact-third Hill response and attenuation operations are known forms; placing local incident intensity inside the response and specifying companion recycling are project hypotheses. The branches are alternatives, not terms to add together. Local filling uses the prescribed full-opacity field. A separate occupancy-dependent-opacity calculation uses $\kappa=k_0 g[1-\eta(XJ)]$ and solves its stationary incident field self-consistently with escaping release. It does not establish formation or support.

For isotropic same-channel return with unit return fraction, a constant normalized intensity $i=J=1$ exactly balances capture and release in the stationary transport equation. This gives the recycling profile in (18). It is an energy-conserving stationary bath with no net bath sink, not a derivation of how an initially empty reservoir forms or persists after illumination stops. Its total spherical stored mass is π squared times $A \eta(X)$ a cubed. Momentum transfer and gravitational support still require their own equations.

With $\kappa=k_0 g$ and isotropic illumination, the effective capture cross section follows from integrating $1-\exp(-\text{optical depth})$ over impact parameter. If available capacity is $A Jg$, the total capacity mass equals (A/k_0) times that cross section. Thus increasing collection area also increases capacity: area alone does not establish a greater energy supply per unit required stored mass. Source luminosities and histories remain needed.

A residual external floor can prolong storage without fixing a universe age. In the selected threshold kinetics, retaining at least 90% of the initial equilibrium occupancy indefinitely requires a remaining intensity between approximately 34% and 69% of the initial intensity across the catalogue. An additive common background large enough to meet that target does not improve the matched galaxy comparison. No observed external companion field has been measured by this inversion.

4.10 Joint lensing and stellar motions

The lens comparison freezes capture amplitudes from galaxy training and then fits stellar mass and constant orbital anisotropy to inner stellar-motion bins. The same mass profile determines bending. Under the two stellar-population proxies, the refined adjusted reference gives lens-angle fractional RMS 13.068%/13.187%; full recycling gives 13.045%/13.161%. The tiny gain occurs in only three of six objects under each proxy and does not repair the rotation or outer-motion discrepancies. The adopted lens angles are catalogue SIE summaries rather than fits to raw lens images; the nonexpanding optical geometry is conditional.

At these fixed motion-fitted stellar masses, stars alone overpredict the required bending in J0037-0942, J1204+0358 and J1402+6321. In the adjusted-reference Chabrier calculation, even removing every companion would require stellar masses lower by at least 12.98%, 12.58% and 15.00%, respectively. This rules out merely subtracting positive companion bending at those fixed stellar fits. It does not rule out a joint change of stellar mass, orbital structure or geometry.

A subsequent diagnostic consumes the lens angle to determine stellar mass, then fits the motions at that mass. Exact lens agreement in this experiment is calibration, not a successful lens prediction. Two distinct nuisance-model extensions were tested separately. One uses a radially varying stellar mass-to-light ratio; the other uses a known smooth anisotropy family [14]:

$$Y(r) = Y_{\text{out}} \left[1 + \frac{h}{1 + (r/R_e)^2} \right]. \quad (19)$$

$$\beta(r) = \beta_0 + (\beta_\infty - \beta_0) \frac{r^2}{r^2 + R_e^2}. \quad (20)$$

Provenance: (19) is a chosen phenomenological gradient, not a derived companion law or a claim of a unique new mathematical form. Equation (20) is a specialization of established orbital-anisotropy models [14]. Beta measures radial versus tangential velocity dispersion; it does not change the deposited density. R_e is fixed to the projected half-light radius. These initially separate extensions are combined in the later diagnostics below; neither changes the shared companion reference law.

The gradient allows h from -0.8 to 9, keeps density positive, and only modestly reduces the large inner-motion cost of the lens-required masses. Increasing the permitted gradient to much larger values in boundary cases still leaves large residuals. For radial anisotropy, initially both endpoints lie between -2 and 0.45. The same three difficult galaxies hit the outer bound. An adaptive follow-up allows their outer endpoint to reach 0.95; all three hit that new bound. Remaining systems retain their earlier interior fits. The improved combined result is therefore a tested configuration, not a demonstrated global optimum.

4.11 Necessary orbital consistency, not established stability

For a spherical separable augmented-density construction with central anisotropy $\beta_0 \leq 1/2$, a nonnegative distribution function must satisfy [15]

$$\gamma(r) = -\frac{d \ln \nu}{d \ln r} \geq 2\beta(r). \quad (21)$$

Provenance: established stellar-dynamical inequality [15], not a companion prediction. ν is stellar tracer density, not total gravitating density. Separability is a candidate completion of the fitted moments, not something those fits established; the criterion is not universal for arbitrary orbital distributions and is not sufficient for positivity.

Using the same multi-component Sersic light profiles, all 12 original-bound configurations and six expanded-bound configurations show no violation on 4097 logarithmically spaced radii from $1e-6$ to $100 R_e$. Minimum margins $\gamma - 2\beta$ are 0.32422 and 0.32546, respectively. Doubling deprojection quadrature changes γ by at most $2.14e-14$; an independent numerical log-density derivative differs by less than $7.27e-5$ away from grid edges. This finite-grid check includes extrapolated light profiles and is not an all-radii proof. It leaves the orbital candidates open but does not construct their distribution functions in the actual potential, demonstrate stability, or alter their remaining motion residuals.

5. Executed results and limits of the evidence

The project record contains both successes and failures. Parameter selection and previously inspected observations must not be described as blind tests. Reported RMS values are descriptive, not likelihood-based evidence for the entire theory.

Test	Current result	What it establishes
SPARC rotation, 149 galaxies	Companion validation/test RMS 32.49/23.59 km/s; MOND 26.88/16.40	Companions improve the ordinary-only control, but tested MOND is better
Six Coma shear bins	Companion/NFW omitted-bin sums 6.86/6.82	Comparable freely fitted shapes; no transferred supply normalization
Six lens galaxies, updated transfer	Adjusted reference 13.07%-13.19%; recycling 13.05%-13.16% lens-angle RMS	Small change, with a material unresolved motion/lensing discrepancy
Earlier imposed Milky Way settling	Inner RMS 3.68/11.83 becomes 2.74/6.70 km/s	A shared conditional inner-fit improvement under two matter baselines
Earlier pilot outer bins	RMS remains 8.36/9.78 km/s	No outer improvement from the selected central rearrangement
Refraction sweep	No selected outer improvement	No reason to promote the tested refraction family
Coupled Milky Way endpoints, 30-60 kpc receiver	All-bin RMS 6.28/8.10 km/s	Conditional angular-momentum and energy accounting, not stability
Finite thresholds, 149 galaxies	12-decade range shifts speeds at most 0.640 km/s	Preserves reference approximately; no improved main residuals
Threshold histories, 149 intensity inputs	First 10% loss takes 4.92%-41.79% of formation time	Conditional kinetics, not observed galaxy ages
Reciprocal protected reservoir	54 stationary cases; balanced energy flows	Replenishment can improve; permanence remains assumed

Test	Current result	What it establishes
Nuclear release	Energy-throughput estimates only	A possible diagnostic channel, not a predicted spectrum or luminosity

SPARC inputs [10] have 89 training, 29 validation and 31 test galaxies in the project split; all have been exposed during development. Companions have three shared fitted constants versus one for the tested MOND law. Restricted shared NFW mappings and per-galaxy NFW fits use different information budgets and cannot be pooled into one ranking. The Milky Way phase pilot uses a spherical gas monopole and two archived stellar baselines, with 20 inner bins for selection and 18 outer bins for comparison. It is not a comparison of two independent galaxies or individual bulge-star orbital predictions [11].

Earlier wave/particle support, saturation, inward migration and coupled exchange tracks remain in the research record. None supplies a demonstrated joint solution for stability, absolute supply and lensing. Numerical mass/energy checks and refinement protect against implementation errors; they do not verify the physical assumptions. The accompanying cross-scale-performance.md and linked executable reports contain the detailed audit trail.

5.1 Matched galaxy comparisons

Each new row below refits only the shared amplitude on the same 89 training galaxies using equal-galaxy velocity mean-square error. Other shared reference parameters remain fixed. This objective differs from the original calibration, so the adjusted reference is the relevant matched control. All partitions have already been inspected. Scores are equal-galaxy RMSE in km/s, not pooled point errors.

Profile or closure	Training	Validation	Test
Adjusted reference	26.87	30.82	21.41
Local filling, prescribed field	28.75	28.63	25.19
Occupancy-dependent transport	29.44	28.64	26.70
Full same-channel recycling	30.77	28.82	30.88
Local finite formation, selected duration	28.55	29.41	25.93

Better validation scores accompanied by worse test scores do not establish a preferred replacement. The finite-formation row selects dimensionless duration 10 from the training scan; it is not a measured galaxy age. None of these rows derives the absolute companion supply.

5.2 Cost of imposing the lens angles

Each entry below gives Chabrier/Salpeter population-proxy results summed across the same six systems. Inner chi-squared uses the measured covariance; outer scores sum squared covariance-conditioned standardized residuals. These are separately reported diagnostics from the earlier inner-bin fits. With the stated full covariance, their sum equals the all-motion chi-squared; this identity is used in the subsequent all-bin analysis. Gradient and anisotropy extensions in this historical table are separate experiments.

Stellar prescription	Inner chi-squared	Outer residual-square sum
Free mass, constant anisotropy	45.81 / 45.27	55.01 / 53.42
Lens-required mass, constant anisotropy	611.48 / 613.61	136.23 / 134.17
Lens-required mass, stellar gradient	471.64 / 476.53	129.03 / 127.53
Lens-required mass, radial anisotropy	447.76 / 453.84	103.09 / 101.47
Expanded outer-anisotropy follow-up	106.88 / 105.62	115.71 / 114.32

The expanded-orbit fit is a substantial inner improvement, but its outer score remains worse than the free-mass control and the previous radial-bound fit. For J1204+0358 and J1402+6321, Chabrier outer standardized residuals remain about 6.98 and 6.54. These conditional residuals are not model-independent exclusion significances. Passing (21) does not remove this discrepancy.

5.3 Combined stellar freedoms and matched controls

Combining equations (19)-(20) initially reduces inner chi-squared to 82.40/81.67, but six of twelve configurations violate (21) near their extrapolated centers. Refitting with the necessary condition enforced gives 87.58/86.79. The adopted light profiles have central tracer slope 0.75, imposing $\beta_0 \leq 0.375$ for the specified completion. This repairs that necessary-condition failure; it does not construct physical orbital distribution functions.

Subsequent fits consume all forty motion bins with their covariance. Releasing the orbital transition radius replaces R_e in (20) by r_a , while the stellar gradient scale in (19) remains R_e . This is the same established anisotropy family [14], with one more fitted stellar parameter. The imposed ranges are h in $[-0.8, 9]$, β_0 in $[-2, 0.375]$ subject also to the sampled slope condition, β_{∞} in $[-2, 0.95]$, and r_a/R_e in $[0.1, 10]$. Central and 8193-point final slope checks pass; positivity and stability remain unproved.

Model	Local fitted parameters per galaxy	Total motion chi-squared
Transferred companion, fixed ra=Re	3	159.83-161.81
Transferred companion, free ra	4	111.27-113.56
Matched stellar-only, free ra	4	175.09
NFW with free scale and strength	6	49.71
Companion with free scale and strength	6	26.49

The population-proxy ranges describe the same six galaxies, not independent repetitions. Stellar normalization is additionally determined using each catalogue lens angle in every row; the count lists parameters optimized against motions. The stellar-only control omits separate gas and central-black-hole components, as do its matched companion comparisons. The transferred profile improves aggregate chi-squared by 35%-36%, but only J0037-0942 and J1402+6321 improve; the other four slightly prefer the stellar-only control. J1402+6321 still contributes approximately 57% of the transferred profile's residual.

Restricted NFW shapes with radius fixed to 0.3, 1 or 3 times the reference capture scale and normalization fixed to the companion bending give 162.62-194.57. Releasing NFW scale and strength reduces this to 49.71. The restricted comparison therefore cannot establish general superiority over dark halos. The NFW density $\rho_s/[x(1+x)^2]$, $x=r/r_s$, is established literature [16], used here as a gravitational comparator without adopting its cosmological formation assumptions. Its ideal untruncated total mass diverges logarithmically and is not treated as a finite source inventory.

5.4 What the freely fitted companion profile establishes

The matched free-companion diagnostic keeps the shared opacity coefficient k_0 fixed, recalculates attenuation for each capture scale a_c , and fits a positive stored-density amplitude D :

$$\rho_d(r) = DJ(r; a_c, k_0)[1 + (r/a_c)^2]^{-2}. \quad (22)$$

Provenance: attenuation and integration are established operations; the opacity shape and its interpretation as deposited companion density are project hypotheses. Equation (22) is a diagnostic relaxation of the reference profile, not a newly derived interaction. The fitted ranges are a_c/Re in $[0.1, 100]$ and companion fraction of required lens bending f in $[0, 0.95]$. For unit-density bending d_{unit} , $D = f \alpha_{\text{required}}/d_{\text{unit}}$; remaining bending calibrates stellar mass. This is a calibration identity, not an energy-source calculation. A free D absorbs the factor $\eta(X)$, so these fits cannot validate the exponent one-third.

Galaxy	Free NFW chi-squared	Free companion chi-squared	Fitted ac/Re
J0037-0942	9.265	4.615	1.963
J1112+0826	17.641	9.060	2.532
J1204+0358	13.945	8.169	0.100
J1402+6321	4.494	0.420	2.078
J1621+3931	2.820	2.762	100.000
J1630+4520	1.548	1.468	0.100

Both free families use 36 local motion-fitted parameters across forty motion measurements, plus consumed lens calibration. Active bounds, shared inherited assumptions and prior inspection of every target prevent interpreting the lower raw chi-squared as independent validation or a complexity-adjusted verdict. Numerical direct-profile polishing and refinement support the score precision, not physical correctness or a global optimum.

The compact J1204+0358 and J1630+4520 fits require density amplitudes approximately 2352-2445 and 493-514 times their respective reference values. These are density ratios, not total-mass multipliers. J1621+3931 instead reaches ac about 877 kpc and implies 8.42e16 solar masses within the numerical grid. Since $J \leq 1$, its positive exterior tail obeys

$$M_d(>R) \leq \frac{4\pi D a_c^4}{R}, \quad E_d = M_d C^2. \quad (23)$$

Provenance: the bound is ordinary integration of $\rho_d \leq D a_c^4 / r^4$. The energy relation is the reference effective-mass closure using established mass-energy equivalence, not a unique local gravitational-field energy definition. The tail adds at most about 1.30% for J1621+3931. Its interior energy equivalent is about 1.51e64 J; no source history supplies this in the current calculation. No universe age or size is imposed to close that deficit.

Crucially, the same galaxy's stellar-only chi-squared is 3.09385, compared with 2.76199 for the enormous free reservoir: a raw improvement of only 0.33186 with added freedom. This endpoint comparison does not show that the data require the enormous mass. The now-executed fixed-scale scan quantifies this tradeoff in Section 5.6. The extreme reservoir is one nearly degenerate fitted branch, not a measured inventory. No free-fit parameters replace the shared exact-third reference.

5.5 Frozen outer predictions and the extended-envelope degeneracy

Without further fitting, the six selected companion profiles have been projected to physical radii from 10 kpc to 10 Mpc. Standard spherical circular dynamics and thin-lens relations [17] give

$$v_c^2(r) = \frac{GM(<r)}{r}, \quad \kappa = \frac{\Sigma}{\Sigma_{\text{crit}}}, \quad g_t = \frac{\bar{\kappa} - \kappa}{1 - \kappa}. \quad (24)$$

Provenance: these are established gravitational relations. Their application assumes the reference effective-density response and the recorded conditional source geometry; it is not a new conversion law. Here $\Sigma_{\text{crit}} = c^2 / [4 \pi G D_l (D_l / D_s)]$ and $\bar{\kappa} = M_{2d} / (\pi R^2 \Sigma_{\text{crit}})$. Circular speed is an ideal orbit diagnostic, not the velocity of every observed star or satellite.

J1621+3931 has negligible extra enclosed mass through the inner few hundred kpc, but almost constant companion convergence near 0.0847 there. Spherical exterior shells exert no interior Newtonian acceleration while contributing to projected lensing. Thus a distant envelope can change bending without comparably changing inner stellar motions. This geometrical degeneracy explains how the large reservoir survives the inner fit; it does not explain how the reservoir forms.

J1621+3931 radius	Implied circular speed (km/s)	Total reduced tangential shear
100 kpc	200.72	0.009849
1 Mpc	63.76	-0.001454
3 Mpc	1520.51	-0.015756
10 Mpc	4236.46	0.034307

The scanned circular-speed maximum is approximately 4246 km/s near 9.17 Mpc. At the adopted lens distance, 3 and 10 Mpc correspond to about 13.7 and 45.6 arcmin. Negative tangential shear denotes radial image stretching, not repulsive gravity. These are extrapolated model predictions without new observed outer velocities or shear. Comparisons must include source distances, foreground structure, environmental mass and nonspherical geometry; a nearly constant convergence is not directly measured by shear alone.

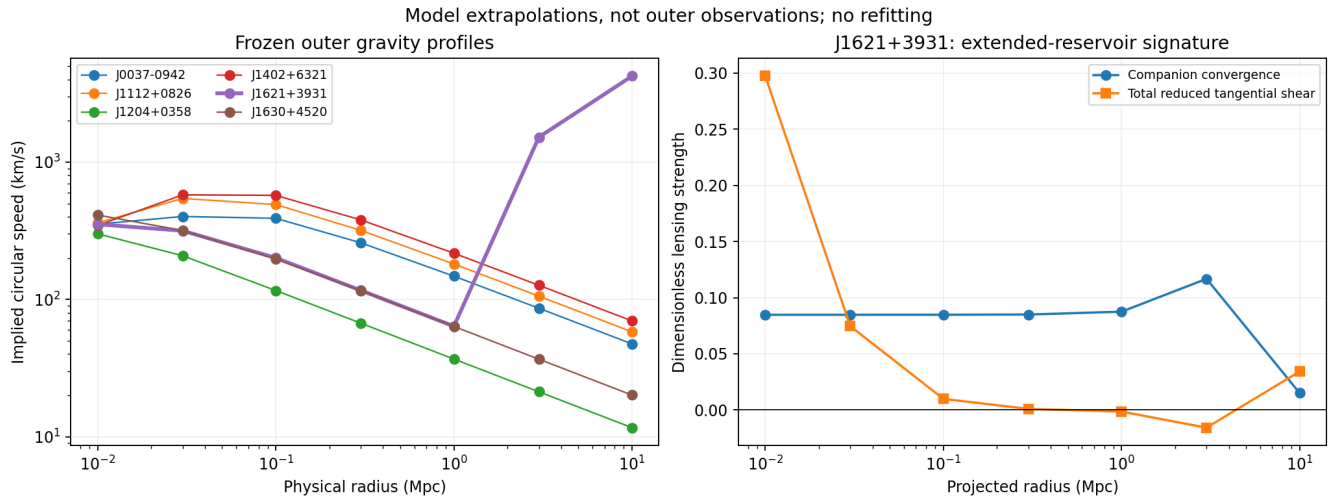


Figure 1. Frozen free-companion outer predictions for six fitted galaxies. Markers are calculated radii; connecting lines guide the eye. No outer observations are shown.

No additional critical curves are found over the finite 0.01-10000 kpc scan beyond the inner radial/tangential pair. J1621+3931 has maximum sampled compactness $2GM/(rc^2)$ about $4.01e-4$ and near-central potential magnitude about $4.79e-4 c^2$. The large mass is diffuse, so its size alone is not an automatic horizon or extra-ring argument. These diagnostics do not establish stability, relativistic completion or the accuracy of isolated thin-lens geometry over this environment.

An initial unsplit angular projection failed a central critical-curve refinement check; splitting at capture-scale crossings resolved the distant-envelope contribution. Final catalogue mean-convergence normalization agrees within 1.55e-6, projected-mass derivative consistency within 3.07e-7 in convergence, and refined critical eigenvalue residuals within 1.22e-8. These are numerical checks rather than measurement uncertainties.

5.6 Updated reservoir assessment: degeneracy and support

Refitting five other local parameters at seven fixed J1621 capture scales changes the interpretation of its extreme mass. The $a_c/R_e=10$ branch has $1.73e14$ solar masses, about 486 times less than the $a_c/R_e=100$ branch, while motion chi-squared changes only from 2.76199 to 2.76284. The stellar-only value is 3.09385. Inner observations therefore do not select a large inventory. These are profile scans of previously fitted data, not confidence limits from a full nuisance-parameter analysis.

Less mass is not simply a weaker version of the same outer field. At 300 kpc the $a_c/R_e=10$ branch predicts circular speed 842.67 km/s and reduced tangential shear 0.03823, versus 116.39 km/s and 0.000984 for $a_c/R_e=100$. The mass has moved inward. Compact and stellar-only branches instead give nearly identical outer predictions. These are frozen extrapolations without new outer observations; environment and source geometry must be included before comparison.

A necessary support test excludes spherical bound collisionless isotropic particles for four of the six free profiles and five of six transferred Chabrier profiles: their deposited density rises outward in an attractive potential. For a nonnegative isotropic distribution $f(\text{relative energy})$, established phase-space integration requires density to increase with relative potential, which decreases outward. This objection concerns deposited particles, not the separately fitted stellar orbits. Passing the condition is not sufficient for a physical distribution function.

An explicit alternative uses randomly oriented circular particle orbits, with no mean rotation but nonzero tangential stress. In the frozen spherical potential,

$$v_c^2 = GM/r, \quad p_r = 0, \quad p_\theta = p_\phi = \rho_d v_c^2/2, \quad \omega_r^2 = G(M + rM')/r^3 > 0. \quad (25)$$

Provenance: established circular dynamics, Jeans balance and an Einstein-cluster-type construction [24], not a new companion interaction. The calculation supports all six free profiles and four nonzero J1621 branches in this singular, fully tangential idealization. Positive ω_r squared establishes individual radial stability in the frozen field, not collective stability or a formation history. The extreme branch has orbital kinetic energy about $1.05e60$ J, or $6.97e-5$ of its effective mass-energy. Binding release may supply this kinetic energy, so it must not automatically be counted as a second external budget. Incoming light-speed companions still need a momentum-exchanging interaction to populate slow bound orbits. Net zero angular momentum does not eliminate individual orbital angular momentum.

The subsequent receiver-recoil calculation keeps capture open kinematically. Under a chosen recoil budget equal to each profile's orbital $K/(M_d c^2)$, a participating receiver rest energy about 7170-677000 times the incident packet energy permits separate-particle production at rest with unchanged receiver invariant mass. These ratios are conditional budget thresholds, not minimum masses required for any capture. Absorption into the receiver's internal state is a distinct branch. The calculations apply established local energy-momentum kinematics [25], not a new interaction law. They do not determine production rates or an accessible particle mass spectrum. A particle created at rest also lacks the angular momentum of the circular support construction. Recoil need not become heat immediately, and ordinary-matter absorption does not

automatically create an extended independent reservoir. This narrows the next question from whether momentum can balance to how a local interaction populates the required bound orbits.

5.7 Newtonian, MOND and dark-halo rotation comparisons

The common 149-galaxy SPARC sample includes gas and uses fixed published distances, inclinations and disk/bulge mass-to-light ratios 0.5/0.7. The original fits minimize equal-galaxy mean squared log10 speed. The following equal-galaxy speed RMSE values are descriptive summaries of those fits, not the fitting objective. There are 89 training, 29 validation and 31 test galaxies, all exposed during development.

Prescription	Shared fitted parameters	Validation RMSE (km/s)	Test RMSE (km/s)
Ordinary-matter Newtonian	0	58.22	47.77
Exact-third companion	3	32.49	23.59
Simple MOND, fitted a0	1	26.88	16.40
Simple MOND, fixed a0	0	26.24	16.49
NFW, imposed shared scaling	3	35.18	40.54

The tested MOND rule [22] and the spherical Newtonian acceleration relation are

$$g_N = GM_b(< r)/r^2, \quad g_{\text{MOND}} = g_b/2 + \sqrt{g_b^2/4 + a_0 g_b}. \quad (26)$$

Provenance: established Newtonian gravity and the known simple MOND interpolation, not our formulas. In the actual disk comparison g_b comes from the supplied gas, disk and bulge rotation contributions rather than a spherical enclosed-mass approximation. Fitted $a_0=8.5633\text{e-}11$ m/s squared; the fixed control uses $1.2\text{e-}10$. This algebraic MOND comparison is not a full nonspherical field solution with external-field effects or a relativistic lensing theory.

The NFW row imposes $r_s=s R_{\text{disk}}$ and $\rho_s=\rho_0 X$ raised to a shared fitted power. Its fitted s reaches the upper bound 100. This restrictive galaxy-to-halo mapping is a project choice, not a general prediction of dark matter. A separate outer-radius experiment fits two halo parameters to each target's inner 60% of radial points:

Model on outer 40% only	Uses target inner motions?	Validation RMSE	Test RMSE
Ordinary matter	No	68.77	61.53
Shared exact-third	No	33.23	29.20
Shared fitted MOND	No	22.14	16.79
Per-target NFW	Yes	16.14	20.07

Units are km/s. These scores use the same 60 comparison galaxies but different radial subsets from the first table. NFW consumes more target information; 41 of 60 fits meet a parameter bound. Outer points were excluded from those local fits but were previously inspected. Neither table supports general superiority over dark matter. Section 5.1 uses a different velocity-space objective and must not be mixed into this ranking.

5.8 Cluster comparison and missing gravitational controls

The Coma diagnostic reconstructs six shear points from Kubo et al. [23], retains the negative outer point and uses the plotted diagonal errors. Its NFW all-bin chi-squared 3.85 approximately reproduces the publication's 3.87, correcting the impression that NFW inherently fits this cluster poorly. No independent source inventory is supplied to the companion normalization.

Fitted shape	All-six chi-squared	Omitted-bin residual-square sum
Transparent companion	3.68	6.86
Strong-interception companion	3.72	10.79
NFW	3.85	6.82
Compact-baryon MOND approximation	4.46	19.62
Point-mass control	7.36	32.17

Each omitted-bin score refits the remaining five bins. This is a small, already exposed shape diagnostic without the original full covariance. Free amplitude absorbs the retention factor and external radiation supply, so it does not test one-third retention. NFW represents a total fitted shape here, rather than a separately measured gas-plus-halo decomposition. The MOND row assumes compact baryons and an imposed equal-potential lensing response, with freely adjusted amplitude and scale; it is not transfer of the galaxy a0 into a realistic extended cluster. The point mass is a numerical control, not a viable ordinary-matter cluster model.

A matched extended-baryon Newtonian/GR and MOND cluster calculation remains absent. A matched MOND analysis of the six lens galaxies is also absent. Their stellar-only lens control uses ordinary stellar gravity with GR weak-field light bending; it does not apply a Newtonian particle-deflection formula to photons. Those missing comparisons prevent claiming that success on galaxies and clusters makes the theory

better than MOND. The six lens galaxies, Coma and the Milky Way pilot have different data, fitted freedoms and score definitions; their raw residuals cannot be added into a global evidence score.

5.9 Conversion rate versus the Hubble constant

Our constant-alpha law has a low-redshift gradient with familiar Hubble units:

$$H_{\text{conv}} \equiv c\alpha = c \frac{dz}{dD} \Big|_{D=0}, \quad z = \exp(H_{\text{conv}} D/c) - 1. \quad (27)$$

Provenance: a definition and the known exponential solution of (1), not a novel expansion equation. H_{conv} measures fractional photon-energy transfer per distance. The actual expansion rate is zero in the stipulated static background; H_{conv} is not that rate and its inverse is not an inferred universe age. No physical theory yet derives its fitted value.

For a specified expansion comparator we use the established flat FLRW distance relations [18], with $\Omega_m=0.3$, $\Omega_{\Lambda}=0.7$ and radiation neglected:

$$D_C(z) = \frac{c}{H_0} \int_0^z \frac{dz'}{\sqrt{0.3(1+z')^3 + 0.7}}, \quad D_L = (1+z)D_C. \quad (28)$$

Provenance: known cosmological relations, used only as a competing explanation. Present-day comoving distance is not the distance a photon physically traveled. The nearby comparison holds the supplied published distances fixed and interprets them alternatively as static path distance or comoving distance; it is a conditional curve comparison, not a reanalysis of the distance measurements under both theories.

Same nearby inputs	Fitted scale (km/s/Mpc)	Eight-region omitted-region RMS (km/s)
Conversion, path	74.618	450.716
FLRW, comoving	74.922	450.502
Conversion, luminosity sensitivity	75.176	450.358
FLRW, luminosity sensitivity	76.038	450.074
Linear distance-redshift control	75.177	450.358

There are 164 recovered galaxy groups at 10.20-93.20 Mpc. Scales use 104 training groups; each eight-region score instead refits on the other seven regions and aggregates predictions across all 164. The old 25-group test RMS is 415.41 versus 414.99 km/s for the first two rows. Paired sky bootstrap 95% intervals for the conversion-minus-expansion RMS difference span zero: approximately [-0.95,1.39] km/s for the first pair and [-2.98,3.58] for the luminosity pair. These diagnostics do not resolve a difference, but they do not establish statistical equivalence or parity on other cosmological observations. Peculiar motions, distance calibration and frame uncertainties remain material.

The luminosity sensitivity row uses stationary photon-number-conserving conversion, $D_L = r \sqrt{1+z}$. It is an alternative interpretation of the supplied distances, not a physical recalibration of surface-brightness-fluctuation measurements. It also differs from the event-stretched brightness branch below. These alternatives must not be combined as one tested model.

For scale, a 100-million-light-year static path gives $z=0.00766048$: a 500 nm photon becomes 503.830 nm and transfers 0.7602% of its energy. Those are predictions of the fitted rule, not a measurement of a particular galaxy. The wavelength change does not by itself stretch an event's duration.

Rate determination	Value (km/s/Mpc)	Meaning and uncertainty
Our nearby H_{conv}	74.62	Training sky-bootstrap 95% range 72.42-76.89; incomplete systematics
Our supernova-brightness H_{conv}	70.48	Formal delta-chi-squared-one range 69.47-71.49; conditional propagation law
Our same-input FLRW H_0	74.92	Fixed 0.3/0.7 comparator above; not a precision external result
Planck base flat LCDM [19]	67.4 +/- 0.5	CMB-inferred H_0 conditional on that cosmology
SH0ES 2022 [20]	73.04 +/- 1.04	Cepheid-supernova distance ladder, reported uncertainty
Local Distance Network 2026 [21]	73.50 +/- 0.81	Covariance-aware combined local distance indicators

The last three rows are published context, not fits to our sample. Their calibrations and assumptions differ, and the local estimates are not independent of each other. Our bootstrap range and formal likelihood interval are not interchangeable one-sigma errors. Numerical proximity to a published H_0 is not evidence for photon conversion. We do not resolve the Hubble tension by renaming or retuning the slope.

The brightness branch assumes event stretching by $1+z$ and a static beam, giving $F=L/[4 \pi D^2 (1+z)^2]$. With 77 calibrator rows, 466 training rows at $0.1 \leq z < 0.3$ and 494 higher-redshift comparison rows, refitting α yields $H_{\text{conv}}=70.475$ and reduces comparison chi-squared from 424.48 to 391.84. Residual means still rise from about 0.062 mag at $z=0.3-0.6$ to 0.225 mag at $z=1-3$. A constant α changes only the magnitude normalization in this branch; it cannot remove that trend. A separately fitted regular beam-area branch reduces the higher-redshift score to 389.18 while retaining a roughly 0.116 mag highest-bin residual. These are exposed-data alternatives, not a common completed law. A matched full FLRW supernova likelihood with identical calibration, covariance and nuisance choices has not been executed here.

5.10 Propagation tests beyond a redshift curve

The following diagnostics test different parts of the proposed propagation mechanism. Each must eventually follow from the same physical law.

Observable	Executed result	Consequence for the common model
Supernova spectral aging, 35 rows	Stationary-duration chi-squared 150.57; 1+z stretch 26.95	Energy loss alone leaves the event-duration problem
Brightness repaired by timing alone	Required exponent 1.481; aging chi-squared 47.06	More delay cannot freely substitute for missing dimming
GW170817 timing	Photon-only affine-time branch predicts about 674646 years extra flight time	That differential-delay branch fails the observed near-coincidence
Limited radio comparisons	Two independent centroid groups, three methanol lines	Common fractional shift is not excluded there; not all-spectrum validation
Microwave spectrum, 43 FIRAS channels	Tested number-retaining thermal history fails	A blackbody origin and transfer law remain to be supplied

The observed GW/gamma arrival separation is 1.74 +/- 0.05 seconds [3], with source emission delay uncertainty. The huge calculated delay belongs to a specific photon-only time law at 40.7 Mpc; it is not a generic consequence of redshift or a measured lag. Equal GW and electromagnetic propagation can cancel their common delay, but that response has not been derived from the energy-transfer mechanism. Constant-speed energy transfer avoids that particular differential-delay calculation while leaving event stretching unexplained. DES reports duration scaling (1+z) raised to b, with b=1.003 +/- 0.005 statistical +/- 0.010 systematic [2]. Source-population assumptions matter, but the duration data remain constraints in the fictional universe. The standard expansion benchmark links wavelength and duration stretching; our proposed mechanism must supply its own common derivation.

There is no executed joint likelihood covering CMB angular structure, baryon acoustic patterns, abundances and all the foregoing tests for our model. Synthetic multiband predictions are not observations. Newtonian gravity alone and a halo density profile alone are not competing cosmological redshift theories, so assigning either a standalone H0 score would misstate the comparison.

6. Evidence that would materially strengthen the theory

The following outcomes retain the full work program. They are not an execution order or claims already achieved. Section 8 selects the next inexpensive discriminating calculations from this broader scope. Each requirement remains open until its own evidence is supplied.

1. A common mechanism with fewer adjustable choices. Specify an interaction and local state equations that yield conversion, capture and support with shared constants. Derive the one-third exponent or replace it with a better justified prediction. Otherwise independent fitting rules can conceal incompatible physics.
2. Absolute supply from independently constrained radiation. Integrate source luminosities, travel, capture, retention and release, keeping age/history as explicit inferred parameters rather than choosing unlimited history to rescue a deficit. Predict effective stored gravity without fitting C_0 separately to the needed mass. Otherwise we have renamed a missing-mass profile rather than explained its origin.
3. Redshift and time stretching from the same propagation law. Use distances independent of redshift, control velocities, predict the same fractional spectral shift across tested bands, and reproduce event durations, sharp images and brightness. Include GW/EM timing. Otherwise matching a distance-redshift curve alone does not explain the observed propagation phenomena.
4. Joint motion and lensing prediction. Use one relativistic response and one deposition profile to predict circular speeds, stellar velocity distributions and strong/weak lensing in the same objects. Include distances, gas, mass-to-light and orbit uncertainty. Otherwise a rotation success cannot repair a lensing failure through an unrelated correction.
5. Genuine transfer to untouched systems. Freeze parameters and analysis choices, then predict additional galaxies and clusters spanning surface brightness, gas content, morphology and environment. Compare residuals, scatter and parameter burden against properly specified MOND and dark-halo models with identical inputs. Otherwise in-sample improvement may reflect flexibility or selection.
6. Formation, stopping and persistence. Solve evolving density, pressure or orbital support, angular momentum and cooling, including feedback from the companion field. Check Solar System and binary constraints in the appropriate limits. Otherwise a useful profile may never form, may collapse, or may disrupt systems already explained well.
7. A radiation prediction distinct from ordinary sources. From the same fixed settling law, predict where energy emerges, its spectrum and variability before using measured brightness. Test both diffuse emission and the optional nuclear channel, including nondetections and jet mechanical power. Otherwise any observed glow can be assigned to companions after the fact.
8. Broader nonexpanding-universe consistency. Address the measured microwave spectrum and angular correlations, large-scale clustering, abundances and thermal histories without importing an expanding-model solution as a premise. Retain observations even when their conventional explanation is set aside. Otherwise a gravitational phenomenology cannot claim to replace a cosmological framework.

7. What would justify a claim of advantage?

A restricted claim is the first realistic target: for example, a frozen companion law predicts joint motions and lensing in a new sample more accurately than specified competitors with comparable or lower effective flexibility. That would be a meaningful result even before every cosmic observable is explained. It should be reported with measurement covariance, nuisance-parameter uncertainty, model complexity and all failed targets included.

A stronger advantage would come from an independently confirmed connection between lost photon energy, accumulated gravity and released radiation. An environmental dependence or radiation feature predicted in

advance, absent from the tested competing models, would be especially valuable. The signature must be quantitatively calculated; an unexplained residual is not automatically companion evidence.

Directly detecting an interaction or state associated with companions would be compelling, but is not the sole route to a viable theory. Conversely, dark-matter direct-detection limits do not automatically promote this theory. Both frameworks must meet the evidence appropriate to their claims. Since companions add an unseen gravitational component, the distinct scientific claim is its radiation-derived origin and calculable dynamics, not simply the absence of unseen matter.

8. Prioritized next steps: reuse existing calculations first

The immediate objective is to resolve the largest ambiguities with small, auditable experiments using the files already available. No new catalog hunt, full cosmological simulation or unrestricted parameter sweep is needed for the first three tasks. All existing samples are exposed: they can diagnose failures and guide development, but cannot become new blind evidence. The order below reflects expected information gained per calculation, not confidence that a branch will succeed. These tasks are proposed, not executed in this revision.

8.1 First: compare the existing propagation fits consistently

Question: does one constant conversion rate explain both nearby redshifts and supernova brightness under one explicitly chosen duration/beam rule? The nearby value 74.62 and conditional brightness value 70.48 already suggest a tension, and changing the constant alone cannot remove the recorded redshift-dependent brightness trend. Another isolated redshift fit would add little.

Smallest useful calculation: reuse direct-conversion/results.json, brightness-distance-consistency/rate-results.json, the archived supernova covariance and calibrators. Evaluate the existing duration-stretched static-beam branch at the nearby rate, at its brightness optimum and at a jointly selected rate. Complete the missing flat-FLRW brightness comparison with the same supernova rows, covariance, calibrator treatment and nuisance freedom, fitting H_0 with the specified 0.3/0.7 background. Distinguish the models' physical distance interpretations; holding a catalogue value fixed does not establish model-independent calibration. Audit shared calibration before adding likelihoods; if its cross-covariance is unavailable, report separate conditional scores and a sensitivity range rather than a precision joint confidence interval.

Deliverable: one compact table of nearby-redshift residuals, brightness residuals by redshift bin, duration consistency and fitted-parameter counts for each candidate. Retain the existing $1+z$ duration assumption as an assumption, not a derived success. Why first: the calculations and data already exist, and this exposes whether a single rate works or a shape/beam change is necessary. Stop condition: if the trend persists, do not keep tuning α ; identify which propagation assumption requires revision. A better score alone does not derive the photon-companion mechanism.

8.2 Second: test whether capture leads to usable orbits

Question: can a conserving local capture prescription populate a long-lived extended reservoir rather than merely stop the incoming energy? The latest recoil calculation removes a kinematic objection, but stopping a particle and putting it on a circular orbit are different achievements.

Smallest useful calculation: reuse the refined ten-profile potentials and angular radiation field. Compare at most three declared injection prescriptions: particles initially at rest; a fixed isotropic slow-velocity distribution; and a tangentially biased distribution with one shared width. These are illustrative proposed prescriptions, not derived microscopic interactions. For each, calculate local binding energy and angular momentum, turning radii and the receiver/outgoing momentum required by conservation. Compare orbit residence with the prescribed density. Use a fixed potential as a first necessary test; do not label it collective stability or self-consistent formation. A circularity distribution that was imposed cannot count as an explanation of its own origin.

Deliverable: bound fraction, radial/tangential motion, recoil budget and predicted redistribution by radius, including which results depend on the chosen receiver mass or injection energy. Why now: capture is kinematically possible, and these inexpensive orbit calculations can expose whether the simplest versions destroy the hollow profiles that helped lensing. Stop condition: retain a branch only as a candidate if it preserves the relevant profile and balances energy/momentum; otherwise name the missing torque, cooling or support process. Do not manufacture a successful profile by assigning a separate injection law to each galaxy.

8.3 Third: locate the shared-law error before adding a parameter

Question: which systems and radii account for the remaining rotation and lensing failures? The shared exact-third law improves ordinary matter but trails the tested MOND rule. Free companion normalization improves lens fits while ceasing to test that law. These are not interchangeable successes.

Smallest useful calculation: reuse saved per-radius SPARC predictions and the six lens residual/covariance records. Without fitting, separate residuals by radius, surface brightness and gas fraction using bins declared before this diagnostic. Report signed errors and uncertainty, not only pooled RMS. Keep the galaxy-rotation objective and lens likelihood distinct, and compare each with its existing ordinary-matter, MOND or NFW controls where actually executed. Identify whether the main failure is normalization, radial shape or a small number of objects. In particular, preserve the fact that only two of six lens systems improve under the transferred profile.

Deliverable: a short residual map and one explicitly motivated candidate modification, or a finding that no shared modification is justified. Why this is inexpensive: the predictions already exist and no new physical parameters are needed for the diagnostic. Stop condition: test at most one additional shared parameter on the existing training subset, retain all comparison outcomes, and reject promotion if improvement comes only from target-specific freedom or damages previously successful limits. Previously inspected validation/test partitions remain development checks. Freeze the resulting rule and protocol before a genuinely untouched-system test.

8.4 Work unlocked by those results

If capture survives the orbit check, calculate a local interaction rate, causal receiver size and cumulative momentum diffusion, then connect radiation supply and retention to the density amplitude. Keep source history free but quantify what history is required; do not use unlimited age to hide a deficit. If the shared gravity residual has a defensible correction, complete the extended-baryon Coma and matched six-galaxy MOND comparisons, recording the chosen lensing completion and matter uncertainties. These are important but require more modeling than the immediate reused-data checks.

Collective stability, a derivation of one-third retention, full Milky Way distribution modeling, emission spectra and the broader microwave/structure/abundance program remain requirements in Section 6.

Companion-powered black-hole emission remains optional; recoil accounting alone predicts neither unavoidable heat nor Hawking-like radiation. Pursue these after the cheaper tests clarify the required mechanism. This ordering changes priority, not the ultimate scope.

Every experiment must state known equations, new assumptions, input provenance, fixed and fitted parameters, numerical checks, failed outcomes and the consequence for the next choice. Keep incompatible propagation and storage branches separate. A useful failed calculation is one that tells us what to change; repeated refinement without changing the physical assessment is not the priority. No background execution is implied by this agenda.

9. Conclusion

The project has established a useful empirical gravity family and a calibrated redshift curve, with numerical checks and explicit failures. Shared companions improve ordinary-matter rotation but do not beat the tested MOND rule. Freely fitted companion profiles improve on matched NFW fits in six lens galaxies, at the cost of removing the shared normalization test; Coma does not distinguish the main fitted shapes. The extreme reservoir is not required, and idealized tangential orbital support is possible, but capture, collective stability and absolute supply remain open. Nearby conversion and expansion curves have almost identical errors, while brightness favors a different conditional conversion rate. A single mechanism has not yet connected these successes to event timing and the remaining observations. The immediate priorities are a consistent existing-data propagation comparison, a conserving capture-to-orbit diagnostic, and a residual map that identifies whether a small shared-law change is justified. Full supply, formation and untouched-system validation remain necessary afterward. These determine whether the model predicts new outcomes rather than assigning a separate rule to each phenomenon. No replacement for MOND, dark matter or expanding cosmology is established.

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Reproducibility and research status

This v1.5 revision reviews evidence through 2929cc8 and prioritizes the next experiments; no new fitting was performed for the manuscript update. The receiver-recoil extension is recorded in `companion-extensions/capture-recoil-threshold.py`, its results JSON and report. Section 8 is a proposed execution order, not additional completed evidence. The comparison audit uses `isotropic-galaxy-transfer/model-comparison-results.json` and `cluster-comparison-detail-results.json`; `direct-conversion/results.json`; `brightness-distance-consistency/rate-results.json` and `regular-area-report.md`; `electromagnetic-audit/report.md`; and `joint-propagation-audit/report.md`. Latest support and degeneracy results are `companion-extensions/reservoir-scale-scan`, `reservoir-branch-outer`, `reservoir-isotropic-support` and `circular-reservoir-support`, each with executable scripts, reports and JSON results. The existing `free-companion`, `free-nfw` and `matched lens-control` records supply the six-galaxy tables. The comparison snapshot and source hashes are recorded in `paper-comparison-audit.json` beside this manuscript. `Cross-scale-performance.md` retains the historical research sequence. The normalization correction remains recorded at 24534de. Alternative population proxies are not extra galaxies; repeated partitions are not new observations. Computational assistance was used in analysis and drafting; independent scientific review remains necessary.